

Computer Network Reference Models

Module 04

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0:00 [Music]

0:16 Hello friends I am Professor yogesh Chaba from Department of computer science and engineering Guru jambheshwar

0:23 University of Science and Technology Hisar harana in this lecture we will

0:30 study about TCP IP reference model first of all please see the contents which I

0:37 shall cover after covering brief introduction about TCP IP reference model I shall take up its layered

0:47 protocol architecture which will be followed by detailed

0:53 explanation about protocols in tcpip reference model

1:01 1 minute, 1 second the Internet Protocol suit commonly known as TCP IP reference model is the

1:08 computer networking model and set of communication protocols used on the

1:15 internet it is commonly known as TCP IP because its most important protocols the

1:24 transmission control protocol TCP and the Internet Protocol protocol IP were

1:32 the first networking protocols defined in this standard the TCP IP reference model is

1:42 the network model used in the current Internet architecture it has its origin back in

1:51 the late 1960s with the grandfather of the internet the

1:58 arpanet this was was a research Network sponsored by the Department of Defense

2:05 in the United States and due to this reason it was also called DOD

2:14 model the major design goals of tcpip reference model

2:20 were ability to connect multiple networks together

2:26 seamlessly ability for connections to remain intact as long as the source and

2:34 destination machines are functioning to be built on flexible

2:43 architecture the TCP IP model and related protocol models are maintained

2:51 by the internet engineering task force ietf the tcpip internet Pro protocol

3:00 suit resulted from research and development conducted by defense

3:07 Advanced research project agency DARPA in the late
3:13 1960s after initiating the pioneering arpanet in
3:19 1969 DARPA started work on a number of other data transmission technologies in
3:28 1972 Robert E KH joined the DARPA information
3:34 processing technology office where he worked on both satellite packet networks
3:42 and groundbased radio packet networks and recognized the value of being able
3:51 to communicate across both in the spring of
3:58 1973 vnon serve the developer of the existing arpanet Network control program
4:06 NCP protocol joined KH to work on open architecture interconnection models with
4:15 the goal of Designing the next protocol generation for the arpanet by the summer
of
4:23 1973 KH and surf had worked out a fundamental reformulation in which the
differences between Network
4:33 protocols were hidden by using a common inter Network protocol and instead of
4:41 the network being responsible for reliability as in the aranet the host
4:49 became responsible the initial four computers connected to the network in the
basic
4:57 arpanet increased to several hundred in just a short time by linking
5:05 universities research institutes and military installations with each other
5:13 over Leed telephone lines as technologically different
5:20 networks such as satellite Network and wireless networks were connected to the
5:26 internet the originally implement ented protocols were soon overburdened by the
data traffic
5:36 translation required from one network to the other now let us see TCP IP reference
5:46 model the actual TCP IP reference model consist of four layers that is layer 2
5:55 to 5 as shown in figure these together together with the addition of the
6:01 6 minutes, 1 second physical layer that is layer one makes up the five layer hybrid
TCP IP
6:09 reference model lower most layer in TCP IP reference model is link layer which
6:17 is also known as host to network layer above link layer is internet layer
6:25 followed by transport layer topmost layer is application layer in
6:34 tcpip reference model the link layer or host to network layer of the TCP IP
6:42 reference model corresponds to the first two layers of the iso OSI reference
6:50 model that is physical layer and data link layer its main task focuses on on
6:59 the secure transmission of data packets of pulled bit sequences it is followed by the
internet
7:09 layer which corresponds to the network layer of the isoi reference
7:17 model its main responsibility is to enable the data communication of Two end

7:24 systems at a given location in the heterogeneous communication network
7:31 the transport layer above it corresponds to the layer of the same name in the ISO
7:39 OSI reference model. It enables two user programs on different computers in the
7:47 communication network to exchange reliable and connection-oriented
7:54 data. The application layer of the TCP/IP reference model includes the three
8:02 upper layers of the ISO OSI reference model and serves as an interface for the
8:10 actual application programs that wish to communicate with each other over the
8:20 network. Unlike the ISO OSI reference model, it was not conceived and planned
8:27 theoretically but was derived from the protocols that had been put into practice on the
8:36 Internet. The ISO protocols, on the other hand, were planned
8:43 theoretically and adopted before protocols could be invented that
8:50 implemented the different functions for the layers of the ISO reference model.
8:59 Today, ISO protocols are no longer used, and the TCP/IP reference model protocols
9:07 which had been developed from practical application dominate the
9:15 Internet. Layered communication in the TCP/IP reference model is as shown in figure
9:24 here. Lower two layers, link and Internet, are chained layers, and upper two layers,
9:33 transport and application, are end-to-end layers where process-to-process
9:41 communication takes place. Now we will see the tasks and the
9:49 protocols of the individual layers of the TCP/IP reference model. First is the link
9:57 layer or Network-to-host layer. The link layer is the lowest layer of the TCP/IP
10:06 reference model. The link layer defines the networking methods within the scope of
10:16 the local network link on which hosts communicate without intervening
10:24 routers. The primary task of the link layer is to ensure your transmission of
10:31 individual data packets between two adjacent end
10:38 systems. This layer of the TCP/IP reference model is normally subdivided into two
further layers: medium access
10:47 control, that is, MAC sublayer, and logical link control, that is, LLC sublayer. MAC
10:56 sublayer regulates access to the shared channels by defining a mechanism of
11:03 allowing fair and efficient access to all participants. This includes methods
11:10 for the discovery of collisions or their avoidance as many participants wish to
11:18 transmit data at the same time. LLC sublayer forms the data link layer of
11:25 the LAN. It provides flow control and link management. Data transmission errors
must
11:34 be recognized and, if possible, corrected. The data is also subdivided
11:42 into size-limited data packets as per requirement. In this layer, important
11:50 protocols of the link layer of the TCP/IP reference model, most important protocols
of the

the link layer of the TCP/IP reference model are based on the IEEE 802 LAN standard these are Technologies such as ethernet IEEE 802.3 token ring IEEE 802.5 as well as different Wireless LAN Technologies IEEE 802.11 and many many more the most important Protocols of the link layer of the TCP/IP protocol family are ATM a synchronous transfer mode ATM which stands for a synchronous transfer mode which is a connection oriented packet switching Network protocol that breaks down and forwards the data to be transported in a cells of a fixed size behind this design principle was the idea of providing time critical realtime data such as video or audio information together with regular data over a standardized protocol ARP address resolution protocol Within the help of the ARP protocol described in RFC 826 the MAC address of a host can be determined from the IP address in the Internet Protocol of the layer above it this is important if a data packet from the Internet is to be delivered to a local network whose Mac address must be determined from the stored IP address of the receiver for forwarding in the LAN

NDP neighbor Discovery protocol the functions of the NDP Protocols are very similar to the ARP protocol and serve to explore and discover further host in the local network in contrast to ARP NDP was developed for the next generation of the Internet Protocol IPv6 while ARP Works under the current version IPv4

link layer topology Discovery the proprietary LLDP protocol was developed by the Microsoft company for exploration of the present Network topology and verification of the guaranteed quality of service in a network PPP Point to point protocol PPP is a simple protocol that serves the connection between two Network nodes PPP is used by the majority of Internet providers to offer their customers a dialup connection over a standard telephone line in the internet second layer in TCP/IP model is internet layer the internet layer has the responsibility of sending packets across potentially multiple networks it requires sending data from The Source Network to the destination Network the Internet Protocol performs three basic functions host addressing and identification which is accomplished with a hierarchical IP addressing system packet routing which is the basic task of sending packets of data from source to destination by forwarding them to the next Network router closer to the Final

16:43 Destination congestion control which is used for control of traffic
16:50 congestion now let us see two basic protocols at internet layer the center
16:58 Cal protocol of the internet layer is the Internet Protocol
17:05 IP IP offers an unreliable data packet oriented endtoend
17:13 information transmission it is responsible for fragmentation and
17:21 defragmentation into IP data grams IP exist today in two versions
17:30 ipv4 and ipv 6 in addition the protocol
17:37 icmp internet control message protocol is implemented in the internet layer it
17:47 is responsible for the notification of specific errors occurring during IP
17:55 transmission as well as for further diagnos G nostic task such as sending
18:02 Eco request to test the availability of a computer and the necessary
18:09 transmission time icmp is a protocol that sits
18:15 directly on the IP two different variations of the icmp protocol exist
18:23 one for ipv4 and another for ipv 6 beside IP and
18:31 icmp there are two more protocols that work on the internet layer of TCP IP
18:39 protocol stack for example IP SEC Internet Protocol security it is
18:47 comprised of a protocol suit to ensure the secure execution of Ip data traffic
18:57 within a data stream IP datagrams can be authenticated and
19:04 encrypted additionally included in IPC are protocols for the handling
19:12 establishment and exchange of secure cryptography Keys called internet key
19:19 exchange protocol Ike igmp internet group management protocol
19:28 protol the igmp protocol carries out the administration of Ip multicast group of
19:37 end systems within a tcpip network special multicast rotors
19:45 administer address list of end systems that can be addressed commonly via one
19:53 multicast address OSF open shortest path
20:00 first the OSF protocol a link state routing protocol transmits IP datagrams
20:10 within a single routing domain or autonomous system OSF is the most widely
20:18 used routing protocol on the internet third layer in TCP IP model is
20:27 transport layer the transport layer establishes a basic
20:33 data channel that an application uses in its task specific data
20:41 exchange the layer establishes process to process connectivity by providing
20:48 endtoend services that are independent of the structure of user data and the
logistics
20:57 of exchanging in information for any particular specific
21:04 purpose the protocol of the transport layer establishes a direct virtual
21:11 endtoend communication connection to allow multiple application
21:18 programs on the same computer every application program is assigned a port

21:25 number to provide unique identification on the transport layer
21:33 every unit of data sent must contain the port number of the sender and the
21:40 receiver in order to be transmitted correctly together with the IP address
21:48 the port number defines what is commonly known as Network socket a unique
21:55 connection end point in the network work endtoend message transmission or
22:02 connecting applications at the transport layer can be categorized as either
22:09 connection oriented which is implemented in TCP or connectionless implemented in
22:18 UDP the transport control protocol TCP is a core element of the Internet
22:25 Protocol architecture and the most most popular protocol of the transport layer
22:32 in the TCP IP reference model standardized as RFC
22:40 793 it carries out a reliable connection oriented bidirectional data exchange
22:48 between two end systems TCP enables the establishment of virtual networks next
22:57 the user data protocol UDP is the second most prominent protocol of the transport
23:05 layer standardized as RFC 768 it transmits independent data units
23:13 known as datagram between application programs that reside on different
23:20 computers in networks the transmission is unreliable possibly combined
23:29 with data loss proliferation of datagrams and changes in sequence the
23:36 datagrams recognized as false are discarded by UDP and do not even reach
23:44 the receiver in comparison to TCP UDP is clearly less complex which is reflected
23:53 in its increased data throughput but UD DP suffers a dramatic
24:00 loss of reliability and security other important Protocols of
24:08 the transport layer are dccp that is datagram congestion control protocol
24:16 which is a message oriented protocol of the transport layer in addition to
24:23 reliably establishing and terminating connections it also distributes overload
24:32 notifications it provides overload or congestion control and can be used for
24:39 the negotiation of transmission parameters RSVP that is resource reservation
24:49 protocol which is used for the request and reservation of network resources
24:57 using IP to transmit data streams it is
25:03 not intended for the actual data transport and bears a similarity to the
25:11 mcomp and igmp protocols at the internet layer RSVP can be implemented by end
25:21 systems as well as routers without having to reserve and maintain specified
25:29 service qualities TLS that is transport layer
25:36 security is predecessor of the secure socket layer protocol
25:43 SSL TLS supplies cryptographic protocols for secure data transport in Internet
25:53 individual TCP segments are encrypted by T LS and
26:01 26 minutes, 1 second SSL TLS provides protocols for the negotiation of
transmission parameters

26:09 for the exchange of cryptographic keys and authentication as well as for encryption
26:18 and digital signature SCP that is stream control
26:25 transmission protocol is a proposal for a highly scalable and performance
26:32 version of the original TCP protocol which is reliable connection oriented
26:40 and is specialized in transmitting large amounts of data next we move on to last
26:50 layer of TCP IP model that is application layer the functions on the
26:56 application layer in the TCP IP reference model can be mapped onto layer
27:03 5 to 7 in the iso OSI reference model
27:10 the application layer primarily functions as an interface for the actual
27:17 application programs wishing to communicate over the network a few of the many
important
27:26 Protocols of the application layer located in the tcpip protocol family are
27:34 tnet that is telnet Network which is defined as RFC
27:41 854 which allows the setup of an interactive bidirectional communication
27:48 connection to a remote computer through a command line
27:55 interface FTP that is file transfer protocol defined by RFC
28:03 959 facilitates the transmission and manipulation of data between two
28:11 computers connected over a tcpip network FTP functions according to
28:18 the client server Paradigm a client initiates the connection and request a service
the
28:27 server takes the connection request and answers the service
28:34 request SMTP that is simple mail transfer protocol defined by RFC
28:42 821 is a simple structured protocol for the transmission of electronic mail on
28:49 the internet HTTP that is hyper text
28:55 transport protocol defined by RFC 2616 is used for data transmission in
29:04 the world wide web just as many other Protocols of the application layer it
29:12 works according to the client server Paradigm and is based on the reliable
transport protocol
29:21 TCP RPC that is remote procedure call defined by RFC
29:29 1057 and RFC 5531 is used for interprocess
29:36 communication that allows a computer program to call an external sub routine
29:43 located in another addressing area DNS that is domain name system
29:52 establishes a name and directory service that delivers the assignment of readable
30:00 and system names to IP addresses for all participating systems on the
30:08 internet SNMP that is simple mail Network management protocol defined in
30:17 341 RFC helps Network Management Systems monitor administer and control
30:27 individual ual systems connected to the network RTP that is realtime transport

30:36 protocol defined by rfc1 189 is used for transmission of real time audio and
30:46 video data over internet so friends this was about TCP
30:53 IP reference model in this lecture we first discussed about introductory
31:00 concepts of TCP IP reference model which was followed by layered protocol
31:08 architecture of tcpip reference model in the end we discussed different protocols
31:17 at different layers of this model thank you
31:28 a
31:35 [Music]